

# **LDBC**

**Collaborative Project**

**FP7 – 317548**

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## **D5.5.4 Second version of Portal**

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***Abstract***

This deliverable reports on the second version of the LDBC Web portal, which is the official portal of the LDBC foundation, where the LDBC press releases, benchmark specifications, community contribution and interaction, and reported (audited) results are going to be hosted. The deliverable provides a detailed report on the organization of the actions to raise requirements, create a corporate web site design and structure of the LDBC Web portal.

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## **Executive summary**

This deliverable reports on the second version of the LDBC Web portal, which is the official portal of the LDBC foundation, where the LDBC press releases, benchmark specifications, community contribution and interaction, and reported (audited) results are going to be hosted. The deliverable provides a detailed report on the organization of the actions to raise requirements, create a corporate web site design and structure of the LDBC Web portal.

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|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Abstract (for dissemination)</b> | This deliverable reports on the second version of the LDBC Web portal, which is the official portal of the LDBC foundation, where the LDBC press releases, benchmark specifications, community contribution and interaction, and reported (audited) results are going to be hosted. The deliverable provides a detailed report on the organization of the actions to raise requirements, create a corporate web site design and structure of the LDBC Web portal. |
| <b>Keywords</b>                     | Dissemination, Portal, Web site, RSS, Blog, Social Media                                                                                                                                                                                                                                                                                                                                                                                                          |

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## **Abbreviations**

CMF – Content Management Framework

CMS – Content Management System

CSS – Cascading Style Sheets

LDBC – Linked Data Benchmark Council

RDF – Resource Description Framework

RSS – Rich Site Summary

XHTML – Extensible HyperText Markup Language

# 1 Introduction

One of the main objectives of LDBC work package 5 (WP5) is to setup and coordinate the dissemination of project activities. All project partners are contributing to this objective that includes a wide range of activities such as the creation and maintenance of the project Web portal, the generation of promotional materials, presentations, posters etc. The portal in particular is a major channel for disseminating the actions of the independent organization, to activate the user community and publish the LDBC results. We aim to make it the *de facto* reference resource for the RDF and graph user community to learn about the performance of the various available systems.

The purpose of this deliverable is to report on the 2<sup>nd</sup> version of the LDBC Web portal that will be used to have an impact on vendors, users and the society in general with the actions undertaken by the LDBC foundation. This is the official portal of the LDBC foundation, where the LDBC press releases, benchmark specifications and reported (audited) results are going to be hosted on one side, and where members of the community, who want to contribute in the graph and RDF benchmarking field, will be able to interact. The deliverable provides a detailed report on the organization of the actions to raise requirements, create a corporate web site design and structure of the LDBC Web portal.

The structure of this deliverable is as follows. In Section 2 we present the actions undertaken by the consortium to secure the appropriate and complete generation of the portal. Section 3 presents the requirements gathered with respect to the LDBC Web portal. Section 4 briefly introduces the design of the LDBC Web portal. Section 5 describes in detail the structure and content of the portal. Finally, Section 6 summarizes the deliverable and briefly describes the future activities with respect to the LDBC Web portal.



## 2 Actions undertaken

Since the first year review meeting on Nov. 26<sup>th</sup> 2013 the consortium has undertaken a set of actions to make sure that the requirements set by the reviewers and the PO are properly addressed. In order to do so, the coordinators of the project (Peter Boncz and Josep L. Larriba) proposed a set of actions that will be coordinated by Ioan Toma and themselves:

1. Hire a company to design the corporate image of the LDBC Foundation and its web page.
2. Make sure that the evolution of the benchmarks (both design and execution of the first benchmarks) progresses as expected to allow their publication in the portal at the time planned. Peter Boncz, as the Scientific Director of the project is responsible for that task.
3. Organize a dissemination team that will be responsible for promoting LDBC after the LDBC portal is deployed. The proposed team includes Serge Tymaniuk from UIBK, Ricard Tapias and Dàmaris Coll from UPC and Iliya Enchev from ONTO. The head of the team will be Dàmaris Coll.
4. Make sure that the portal is fully developed. Serge Tymaniuk from UIBK is responsible for this task.
5. Set a number of regular meetings with the people responsible for each aspect to coordinate the development of the Portal and the successful completion of dissemination tasks. The meetings are coordinated by Josep L. Larriba Pey.

Those actions were discussed and a plan to implement them was agreed during a General Assembly teleconference of the consortium on March 6<sup>th</sup> 2014

The following persons will be acting in the roles specified above with the following responsibilities:

1. Berta Hernández. She is responsible for design the corporate image of the LDBC Foundation and its web page.
2. Serge Tymaniuk. He will be the Web master, developing the portal.
3. Dàmaris Coll. She will coordinate the dissemination team and provide insight in the design of the corporate image and web portal.
4. Ricard Tapias. He will take care of the promoting the portal and disseminating the benchmarking results among the Graph community.
5. Iliya Enchev. He will take care of the promoting the portal and disseminating the benchmarking results among the RDF community.

With this in mind, the coordinators have prepared a set of actions that will be explained in the Dissemination deliverable D5.5.2 by month 24.

### 3 Requirements

The development of the second version of the LDBC Web portal was driven by a set of requirements that were identified by the LDBC consortium members. This section lists and explains these requirements. Each requirement is described using one of the keywords "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL".

1. MUST, REQUIRED, SHALL - The *requirement* is an absolute requirement. The final LDBC foundation web site *must address this requirement*.
2. SHOULD, RECOMMENDED - There may exist valid reasons to ignore *this requirement*, but the implications of *doing so* must be understood and weighed before doing so.
3. MAY, OPTIONAL - The *requirement* is truly optional. The final LDBC foundation web site may choose to omit the *requirement for the sake of scope*.

The requirements are grouped in three main categories namely: content, usability and user experience, and social media presence. In the following sections we describe how these requirements were addressed.

#### 3.1 Content

| Req. nr. | Req. description                                                                                                                                                                             |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ_C1   | The homepage <b>MUST</b> contain a list of the topics covered. It <b>MUST</b> discuss the character, objective, etc. of LDBC Foundation.                                                     |
| REQ_C2   | The homepage <b>MUST</b> contain the most relevant benchmarks with links to their subpages                                                                                                   |
| REQ_C3   | The homepage <b>MUST</b> contain news of the LDBC Foundation                                                                                                                                 |
| REQ_C4   | The homepage <b>MUST</b> contain a list of the web contents with hyperlinks to the desired web page. The contents include: (i) the benchmarks, (ii) rules, (iii) results for the benchmarks. |
| REQ_C5   | The website <b>MUST</b> highlight the downloads and the benchmarking software, that will be published on github                                                                              |
| REQ_C6   | The website <b>MUST</b> include widgets that show download counts and github activity They <b>SHOULD</b> be very visible.                                                                    |
| REQ_C7   | The website <b>MUST</b> include videos and webinars prominently to engage and to let the LDBC story be heard.                                                                                |
| REQ_C8   | The website <b>MUST</b> include a "Useful links" info.                                                                                                                                       |
| REQ_C9   | The website <b>MUST</b> include a "Who we are" info.                                                                                                                                         |
| REQ_C10  | The website <b>MUST</b> include a "What's new" info.                                                                                                                                         |
| REQ_C11  | Useful links, "Who we are", "What's new", <b>MUST</b> be in a different place than the information about the benchmarks, rules, results.                                                     |
| REQ_C12  | The website <b>MUST</b> contain an updated selection of papers about benchmarking, classified among those that use LDBC workloads, and those that use other workloads.                       |
| REQ_C13  | The website <b>MUST</b> include links to the conferences related to LDBC.                                                                                                                    |
| REQ_C14  | The website <b>MUST</b> have an intranet section accessible only to the members of LDBC Foundation.                                                                                          |
| REQ_C15  | Download links <b>MUST</b> be provided for every major LDBC documents (.doc, .pdf)                                                                                                           |
| REQ_C16  | Every released document <b>SHOULD</b> be versioned, ideally with automated diffs between them.                                                                                               |
| REQ_C17  | The website <b>MUST</b> include a blog section.                                                                                                                                              |

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ_C18 | The website <b>MUST</b> have a section describing the core LDBC concepts. Content for this section comes from the blogs.                                                                                                                                                                                                                                                                                                                                                |
| REQ_C19 | Each benchmark <b>MUST</b> have its own page which includes: (a) "story" about the use case and why it was selected, (b) a description of the data generator or dataset, (c) description of the employed schemas, (d) description of the queries including purpose/challenges of each query (ideally in Choke Point style), (e) explanation of parameter substitution, (f) presentation of results for the benchmark and for specific engines that used it for testing. |
| REQ_C20 | It <b>MUST</b> be clear from the page the system features that this benchmark tests, and also suggest other benchmarks that test the same or similar features (cross-referencing)                                                                                                                                                                                                                                                                                       |
| REQ_C21 | The content on benchmarks and results <b>SHOULD</b> follow the structure on the TPC website ( <a href="http://www.tpc.org/">http://www.tpc.org/</a> )                                                                                                                                                                                                                                                                                                                   |
| REQ_C22 | The content on specific benchmark executions and results is large in volume, so it <b>MUST</b> be generated from machine-readable artefacts. These <b>SHOULD</b> be in RDF and backed by an ontology (aka W3C EARL format for conformance test results)                                                                                                                                                                                                                 |

### 3.2 Usability and user experience

| Req. nr. | Req. description                                                                                                                                              |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ_U1   | The website <b>MUST</b> have a modern look & feel similar to modern "web2.0" companies (every page fits on one screen, first page is sparse, menus are light) |
| REQ_U2   | Information <b>SHOULD</b> be presented in a visually appealing web format                                                                                     |
| REQ_U3   | Information about benchmarks <b>SHOULD</b> be presented in a way that is simple, attractive, and concise.                                                     |
| REQ_U4   | Access to individual benchmarks <b>MUST</b> be quick (at most two clicks) and easily visible.                                                                 |
| REQ_U5   | For each benchmark the description of its dataset, both schema and distributions, <b>MUST</b> be both in text and visual formats                              |

### 3.3 Social media presence

| Req. nr. | Req. description                                                                                                                                   |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ_S1   | The website <b>MUST</b> include a Twitter feed with all the tweets about graph, RDF and benchmarking that appear                                   |
| REQ_S2   | The website <b>MUST</b> include a summary of recent tweets, Facebook comments, LinkedIn conversations, blog posts, articles by/about LDBC, videos. |

## 4 Design

The design of the corporate image and web page has been subcontracted to a professional designer, Berta Hernández from Barcelona. Both aspects have been taken anew by this person who has given a fresh view to all the concepts.

The design of the corporate image has been done anew and the different aspects of the LDBC logo and FP7 project image have been taken into account to create the new image. The creation of this corporate image has provided a logo itself with a significant relationship with the old logo, both in terms of shape and graph content, making sure that the colors used are significant for an industrial organization.



Regarding the web page, the use of one of the templates of Drupal as the basic template for the design has resulted to a significant reduction in the amount of time required for the design and implementation of the web site. This can be seen as the main characteristic, and the interaction/collaboration between Berta Hernández and Serge Tymaniuk have allowed for a fast implementation of the proposed designs.

## 5 LDBC Web portal

In this section we give a short overview of the content, structure, and supporting technologies used to build the LDBC Web portal. The portal is temporarily hosted at <http://ldbccouncil.sti2.at>. The permanent location of the portal is <http://ldbccouncil.org> where the portal will be soon available.

The LDBC Web portal is the entry point for getting an overview of the LDBC foundation and its mission towards building the reference place for benchmarking of graph and RDF data management systems. The structure and content of the LDBC Web portal was created to address the requirements defined in Section 2. The LDBC Web portal aims to provide easy and fast access to all interested parties (including graph and RDF industry, developers, user community) to specific information about benchmarks and newest results and happenings within and outside the LDBC foundation. The structure is quite flat, with only three levels at most, so that in a couple of clicks from the home page, anyone may reach the information they want. The LDBC Web portal is periodically updated to reflect the latest advancements with respect to the development of the LDBC benchmarks and related activities.

The LDBC Web portal contains the following areas:

1. **Home:** This is the starting point for graph and RDF industry, developers, users and all interested parties, and therefore contains a summary of all LDBC Foundation information. The home page gives a quick overview of the LDBC foundation having dedicated entry points for:
  - o **Industry**, providing a short overview about benchmarks whose target audience are industrial vendors.
  - o **Why LDBC?** containing the reasons for LDBC to exist.
  - o **Community**, providing various ways for the community to express their thoughts about the benchmarks, to propose changes, to present related results like papers on benchmarks for graph databases and RDF engines etc.
  - o **Blog**, containing a summary of the latest blog entries with a pointer to the blog section.
  - o **Contact**, containing the contact information.

The following requirements identified in Section 3 are addressed in the homepage of the portal: REQ\_C1, REQ\_C2, REQ\_C3, REQ\_C4, REQ\_C10, REQ\_C11, REQ\_U1, REQ\_U2, REQ\_S1, and REQ\_S2.

The home page of the LDBC Web portal can be seen in Figure 1.



Figure 1: LDBC Web portal - Homepage

2. **Industry:** This area provides a general LDBC benchmarking overview. It contains the most relevant results for the last benchmark executions accepted by the auditors, and an index of all the benchmarks, with a short description. This area contains further subsections, namely:
  - o **Results**, showing all the results of the benchmarks sorted by type of benchmark.



- o **Benchmarks**, listing the different benchmarks, with a link to the document describing them, and the software repositories namely Github, where to find the software to generate the data and to execute the benchmark workloads.
- o **Intranet**, the area accessible only to members allowing them to access a special set of features, like: task forces where they are subscribed
- o **Members**, showing the logos and pointers to all the members of the LDBC foundation.

The following requirements identified in Section 3 are addressed in this section of the portal: REQ\_C14, REQ\_C15, REQ\_U3, and REQ\_U4. Updates of this section will address REQ\_7, REQ\_C16, REQ\_C19, REQ\_C20, REQ\_C21, REQ\_C22, and REQ\_U5.

The **Industry** section of the LDBC Web portal can be seen in Figure 3.

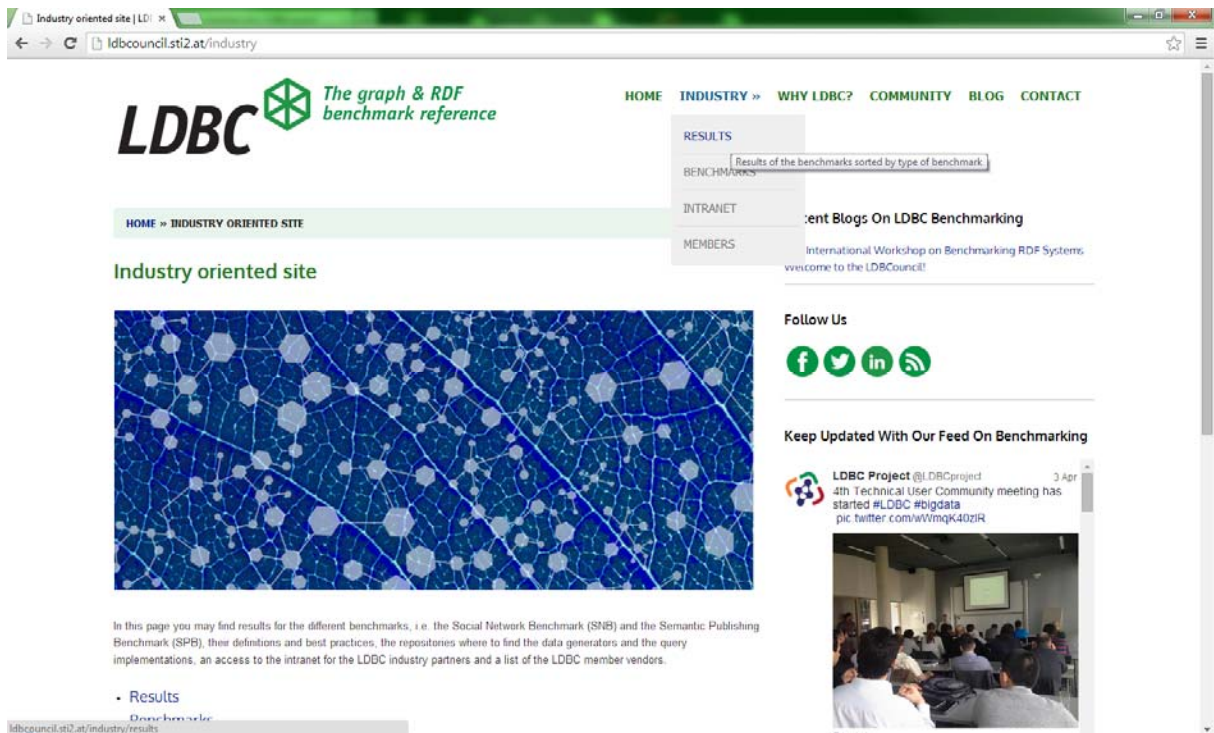


Figure 2: LDBC Web portal – Industry section

3. **Why LDBC?**: This section provides information about
  - o **What is a benchmark?**, explaining why they came to life, for which use case, what are their objectives,
  - o **A short history of benchmarking**, summarizing the most important benchmark efforts, especially in the area of data management,
  - o **What is LDBC?**, explaining the exact structure of LDBC: task forces, benchmarks, planned benchmarks for the future, a brief explanation of the results, the metrics, etc.,
  - o **Other benchmarking efforts**, pointing to other existing benchmarking efforts,
  - o **History of LDBC**, describing who formed LDBC originally, who funded it and the reason for LDBC to come alive.

The following requirements identified in Section 3 are addressed in this section of the portal: REQ\_C9.

The **Why LDBC?** section of the LDBC Web portal can be seen in Figure 3.

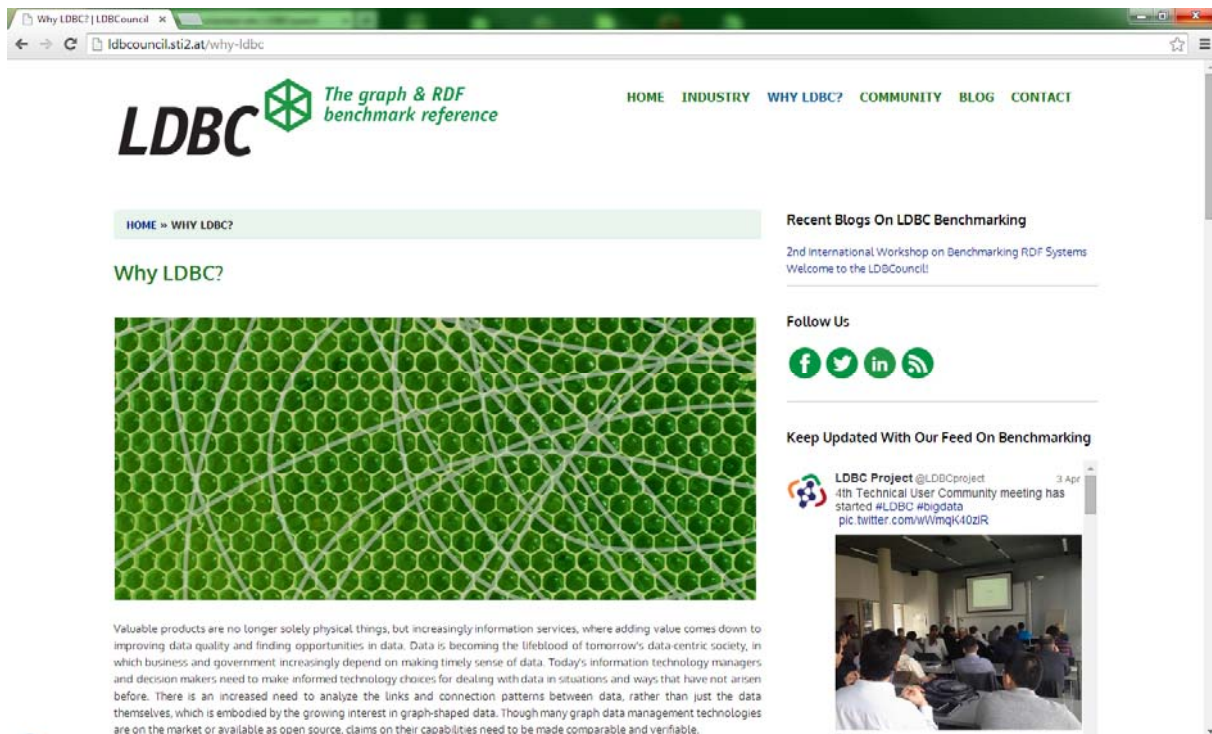


Figure 3: LDBC Web portal – Why LDBC section

4. **Community:** This section provides various ways for the community to express their thoughts about the benchmarks, to propose changes, to present related results like papers on benchmarks for graph databases, etc. This section includes the following subsections:
  - o **Papers on benchmarking**, containing links to papers on benchmarking for graph and RDF databases,
  - o **Moderated discussion board**, providing a proper discussion area i.e. moderated forum
  - o **Links to the Github repositories for each benchmark**, containing clear explanations of the different parts of the benchmarks and where to find each,
  - o **Rules to change the benchmarks**,
  - o **Links to conferences on the topic**, containing links to all the relevant conferences related to RDF and graph benchmarking.

The following requirements identified in Section 3 are addressed in this section of the portal: REQ\_C5, REQ\_C8, REQ\_C12, and REQ\_C13. Updates of this Web portal section will address REQ\_C6.

The **Community** section of the LDBC Web portal can be seen in Figure 4.



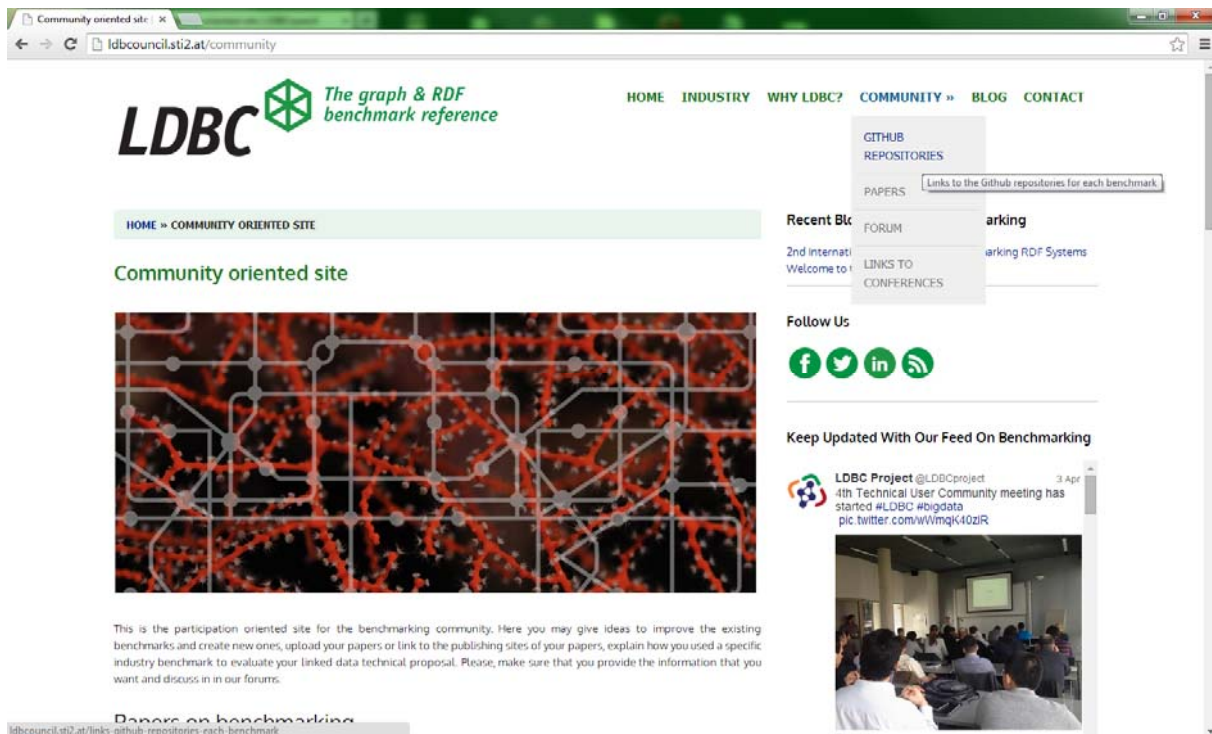


Figure 4: LDBC Web portal – Community section

5. **Blog:** This section contains the LDBC blog posts.

The following requirements identified in Section 3 are addressed in this section of the portal: REQ\_C17. Updates of this Web portal section will address REQ\_18. The **Blog** section of the LDBC Web portal can be seen in Figure 5.

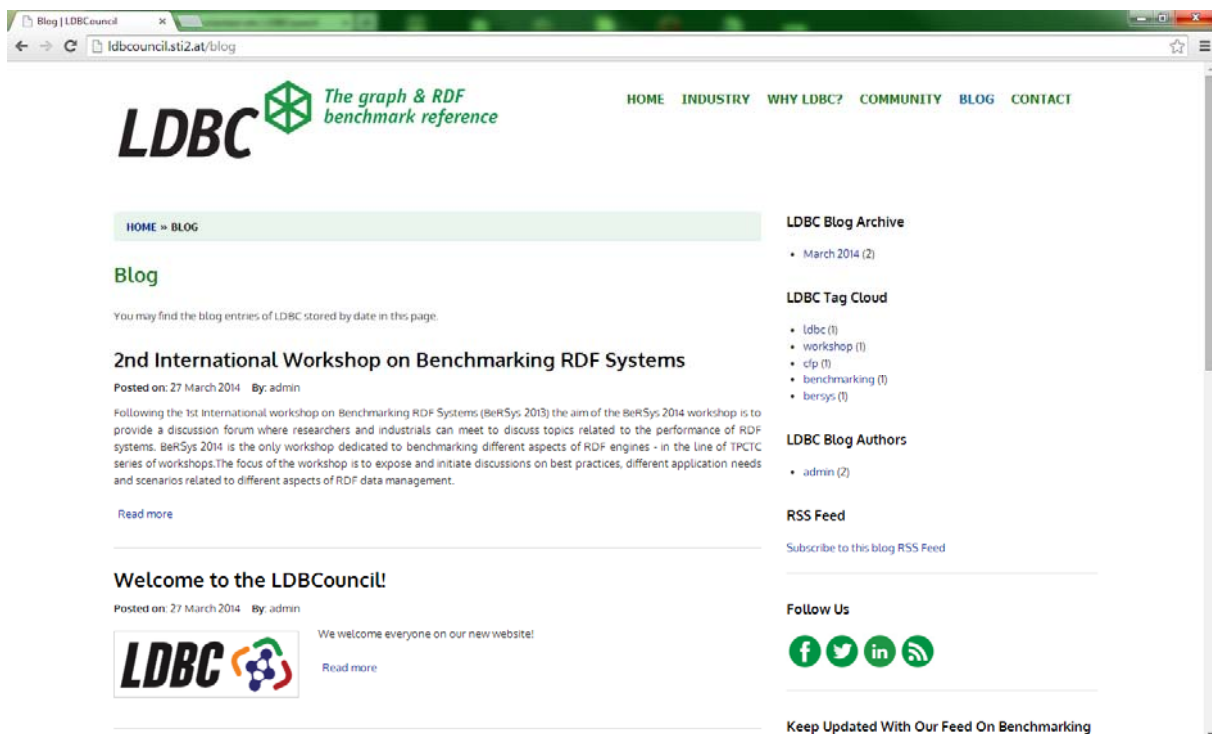


Figure 5: LDBC Web portal – Blog section

6. **Contact:** This page contains contact information.

Additionally, we make the latest news available as an RSS feed, and disseminate LDBC on various social media channels including Twitter, Facebook and LinkedIn (see Figure 6).

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#### Follow Us



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#### Keep Updated With Our Feed On Benchmarking



Figure 6: LDBC Web portal – Social Media

## 5.1 Technologies

The LDBC Web portal is based on Drupal 7. Drupal<sup>1</sup> is a free and open source content management system (CMS) and content management framework (CMF) written in PHP and distributed under the GNU General Public License. It is a very popular platform with a very large end-user and developer community. Drupal is in full compliance with W3C Web standards to allow interoperability with different browsers. The design of the LDBC Web portal was created using Photoshop as described in Section 4 and was afterwards transferred to Drupal by building a theme from the original image with respect to CSS2<sup>2</sup> and XHTML 1.0<sup>3</sup> validity. As a basic theme to setup the portal we used the Nexus theme<sup>4</sup>, a responsive, mobile-friendly theme. The LDBC Web portal seen on a mobile device is displayed in Figure 7. The LDBC Web portal is a HTML5 website<sup>5</sup> which includes a flexible image slideshow with caption and multi-level drop-down menus.

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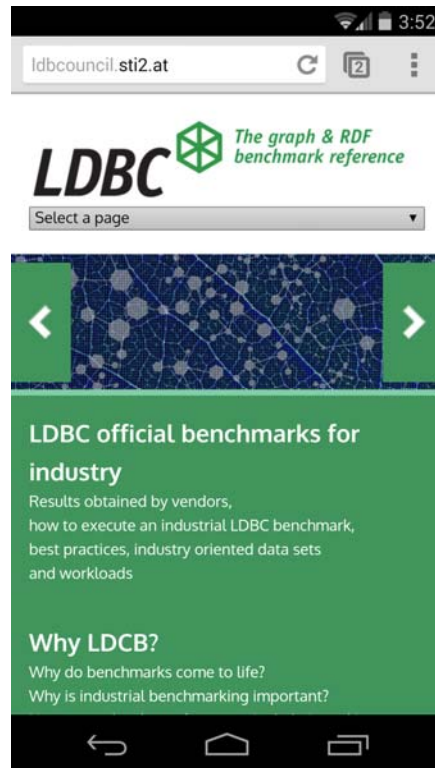
<sup>1</sup> <http://drupal.org/>

<sup>2</sup> <http://www.w3.org/TR/1998/REC-CSS2-19980512/>

<sup>3</sup> <http://www.w3.org/TR/xhtml1/>

<sup>4</sup> <https://drupal.org/project/nexus>

<sup>5</sup> <http://en.wikipedia.org/wiki/HTML5>



**Figure 7: LDBC Web portal – Mobile version**

## 6 Conclusions

This deliverable gives an overview of the 2<sup>nd</sup> version of the LDBC Web Portal. We describe the LDBC Web portal design, structure and setup. The website is the official portal of the LDBC foundation, where the LDBC press releases, benchmark specifications and reported (audited) results are going to be hosted. With the basic structure and content in place, as described above, we will populate the portal with more information about benchmarks, results, etc. as they become available and we will use the portal as a central point to engage, build and grow the LDBC community.